

# AES 41

## Abstract

AES41 provides for the carriage of audio meta-data by embedding it in the audio samples themselves. This tightly associates the meta-data with the audio, yet makes it fragile so that changes to the audio will invalidate the meta-data. Several meta-data sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This Part specifies the method by which a meta-data set is embedded in the audio samples. Other parts define the content and format of the meta-data sets.

## PART 1

### 0 Introduction

#### 0.1 Rationale for this standard

It has become apparent that there are requirements for the transport of audio meta-data together with the related audio, in a way such that the two are not separated. Further, there are requirements where any alteration of the audio should render the related meta-data invalid. Meta-data held in files, or transported in associated video signals, or in AES3 user bits, for example, does not meet these requirements: The meta-data may become dissociated from the audio, or may continue to be associated with the audio even after substantial changes have been made.

Initially, this standard was developed to satisfy these requirements, of tight association and fragility, for an application in cascaded digital compression (bit rate reduction).

Other applications have been identified that have these same two requirements, including the carriage of meta-data to describe loudness.

### 3.2 Symbols

The mathematical operators used in this standard are similar to those used in the C programming language. The bitwise operators are defined assuming 2's-complement representation of integers. Numbering and counting loops generally begin from zero.

#### 3.2.1 Arithmetic operators

##### 3.2.1.1

+

addition

##### 3.2.1.2

++

increment

##### 3.2.1.3

--

decrement

3.2.1.4

\*

multiplication

3.2.1.5

%

modulus operator, defined only for positive numbers

### **3.2.2 Relational operators**

3.2.2.1

<

less than

3.2.2.2

==

equal to

3.2.2.3

!=

not equal to

### **3.2.3 Bitwise operators**

&

AND

### **3.2.4 Assignment**

=

assignment operator

## **3.3 Abbreviations**

The following mnemonics are defined to describe the different data types used in the coded bit stream.

### **3.3.1 bslbf**

bit string, left bit first, where "left" is the order in which bit strings are written in ISO/IEC 13818

NOTE Bit strings are written as a string of logic 1 s and logic 0 s within single quote marks, for example, 1000 0001 . Blanks within a bit string are for ease of reading and have no significance.

### **3.3.2 rpchof**

remainder polynomial coefficients, highest order first

### **3.3.3 simsbf**

signed integer, most significant bit first

### 3.3.4 uimbsf

unsigned integer, most significant bit first

## 7 Semantics of the bit-stream syntax [remove “coder control” wording]

The **data\_frame** syntax shall be as described in this clause.

**header** Part of the bit stream containing synchronization and content indicators.

**data** Part of the bit stream containing information about the audio signal (for example, previous encoding, loudness information).

**error\_check** Part of the bit stream containing information for error detection.

**ancillary\_data** Part of the bit stream that may be used for ancillary data.

### 7.1 Header

The first 54 bits constitute the header information which is common to all ~~audio-coder-control~~ data types, where:

**synchronization\_word** is the bit string ‘0000 0000 0000 0000 0000 0001 1111 1011’;

**length** is the number of bytes (octets) in the recoding\_data\_frame following the length count, up to and including the last bit of the CRC, whereby *padding\_bits* in the *data* ensure that the length is a multiple of 8 bits;

**type** indicates the type of data that follow. See Table 1;

**flags** are bits indicating the validity of fields in the audio coder control data frame header;

**introduced\_time\_offset\_flag** is a 1-bit flag, which when set to logic 1 indicates that the introduced\_time\_offset field in the header is valid; when set to logic 0 it indicates that the introduced time offset field is not valid (in which case the introduced\_time\_offset should be set to logic 0);

**introduced\_time\_offset** indicates a time offset, measured in sample periods, by which the audio signal has been shifted with respect to a reference time. A positive offset represents a delay, a negative offset indicates that the audio signal is in advance of the reference time. The time offset is represented by a signed integer with the most significant bit inverted to reduce the probability of imitation of the synchronization word.

Table 1 - Meta-data types

| Type | Meta data                                            |
|------|------------------------------------------------------|
| 0    | Undefined (see note)                                 |
| 1-29 | Defined by other parts of this standard, or reserved |
| 30   | Reserved                                             |
| 31   | Reserved                                             |

NOTE The type logic 0 (undefined) should be used in situations where there is no knowledge of a previous encoding scheme, including those where the audio has not yet been encoded, or this transport mechanism is being used to carry data that is not one of the standardised types.

### 7.3 Syntax for error\_check

Where:

**crc\_check** is a 16-bit cyclic redundancy check word used for optional error detection within the audio coder control data stream.

This field is byte-aligned with respect to the start of the *header*. The bits included in the CRC check are all those in the *header* and *data* sections. The generator polynomial for this check word is:

$$G(X) = X^{16} + X^{15} + X^2 + 1$$

The method is depicted in Figure 1. The initial state of the shift register T should be ‘1111 1111 1111 1111’. All the bits to be included in the CRC are shifted into the shift register in the order in which they occur in the audio coder control data bit stream. After the last shift operation, the outputs b15...b0 constitute the CRC word to be used.

Data bits

T15 b15 T14 b14 T13 b13 T2 b2 T1 b1

Figure 1 — CRC check generator

### 7.4 Syntax for ancillary\_data

Where:

**ancillary\_bit** carries optional information conveyed according to a user-defined format

The number of ancillary bits equals the total number of bits that can be transmitted during one audio frame minus the number of bits in the *data\_frame*.

The minimum *data\_frame* length is the length of the header plus the length of the *error\_check*. This would be the case when the meta-data type is set to logic 0 (undefined).

## 8 Method of signaling the meta-data bit stream on the AES3 interface

Three signaling methods are defined for use with the AES3 interface, according to the resolution (bits/sample) of the systems that are to be interconnected. The methods signal the data bit stream by altering the parity of the audio samples. Even parity indicates a data bit with value logic 0, odd parity indicates a value of logic 1.

If an audio channel can only transport the sixteen most significant bits of the AES3 sample value, then the data should be carried by having the sixteenth most significant bit set so that the parity of the sixteen most significant bits is as required.

If the audio channel can transport twenty or more bits transparently, then the data should be signaled by having the twentieth most significant bit set so that the parity of the twenty most significant bits is as required.

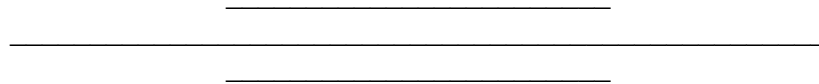
If the audio channel can transport twenty four or more bits transparently, then the data should be signaled by having the twenty-fourth most significant bit set so that the parity of the twenty-four most significant bits is as required.

If a monophonic audio signal is being carried on one channel of an AES3 interface, the data should

be signaled in the same channel as the audio signal is being carried.

If an audio signal consisting of two channels (either stereophonic or independent) is being transmitted on an AES3 interface, separate data should be signaled for the two channels. The data for each channel should be sent in the channel in which the signal to which it relates is being transmitted.

Channel status data shall indicate that the interface is being used for audio.



# AES 41

## PART 2 MPEG-1 and MPEG-2 LSF Layer II meta-data

### 0 Introduction

AES41 provides for the carriage of audio meta-data by embedding it in the audio samples themselves. This tightly associates the meta-data with the audio, yet makes it fragile so that changes to the audio will invalidate the meta-data. Several meta-data sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.

This part describes the format for the data to be transmitted with audio decoded from MPEG-1 Layer II or MPEG-2 LSF Layer II encoding. The data may be used to aid a subsequent re-encoding process. The presence of this data in the stream is indicated by **type**, as defined in Part 1 of this standard, equal to 2.

### 0.1 Rationale for part 2 of this standard.

Digital compression techniques allow the use of lower bandwidth channels and less storage space or, conversely, allow faster transmission and the storage of more program material. Digital compression will be used for a significant amount of program acquisition, for program production and storage, and for distribution and broadcasting.

Compression systems add impairments to sound. The audibility of these impairments depends on the degree of compression and the techniques used.

It has been proved that re-encoding with minimal impairment can be achieved by using information describing the previous encoding process. The information that is preserved varies according to the encoding scheme that has been used, but might include framing and quantization resolution data.

## 3[should be 2] Definitions, symbols, and abbreviations

### 3.1 Definitions

#### 3.1.1 analysis filter bank

filter bank in the encoder that transforms a broadband pulse-code-modulation (PCM) audio signal into a set of subsampled sub-band samples

#### 3.1.2 channel

means of carrying an audio signal (for example, a sequence of data forming part of a bit stream)

#### 3.1.3 coded-audio bit stream

coded representation of an audio signal

#### 3.1.4 compression

reduction in the number of bits used to represent an item of data

#### 3.1.5 CRC

cyclic redundancy check

#### 3.1.6 dual-channel mode

mode where two audio channels with independent program contents (for example, bilingual) are encoded within one bit stream

NOTE The coding process is the same as for the stereo mode.

#### **3.1.7 emphasis**

filtering applied to an audio signal before storage or transmission (for example, to improve the signal-to-noise ratio at high frequencies)

#### **3.1.8 joint stereo coding**

any method that exploits stereophonic irrelevance or stereophonic redundancy

#### **3.1.9 joint stereo mode**

mode of the audio coding algorithm using joint stereo coding

#### **3.1.10 layer**

one of the levels in the coding hierarchy of the audio system defined in ISO/IEC 11172-3

#### **3.1.11 multichannel**

combination of audio channels conveying the audio signals used to create a spatial sound field

#### **~~3.1.12 nextbits function~~**

~~nextbits()~~

~~function that permits comparison of a bit string with the next bits to be decoded in the bit stream~~

#### **3.1.13 scale factor**

factor by which a set of values is scaled before quantization

#### **3.1.14 scale-factor index**

numerical code for a scale factor

#### **3.1.15 stereo mode**

mode where two audio channels which form a stereophonic pair (left and right) are encoded within one bit stream

NOTE The coding process is the same as for the dual-channel mode.

#### **3.1.16 sub-band**

subdivision of the audio frequency band

#### **3.1.17 sub-band filter bank**

set of band filters covering the entire audio frequency range

NOTE In ISO/IEC 11172-3, the sub-band filter bank is a polyphase filter bank.

#### **3.1.18 sub-band samples**

filtered samples created by the sub-band filter bank within the audio encoder, constituting a filtered and subsampled representation of the input audio stream

NOTE From 384 time-consecutive input audio samples, twelve time-consecutive sub-band samples are generated within each of the 32 sub-bands.

#### **3.1.19 synthesis filter bank**

filter bank in the decoder that reconstructs a PCM audio signal from sub-band samples

### **3 Information for transparency**

In order to re-encode decoded audio with minimal impairment, it is necessary to re-encode using the same coding decisions that were used by the previous generation (upstream) encoder. For example, for near-transparent cascaded coding of MPEG-1 Layer II or MPEG-2 Layer II audio, the information needed by the downstream encoder includes:

- . the positions of the frame boundaries;
- . the bit rate of the compressed bit stream before decoding;
- . the coding mode (monophonic, joint stereo, and so on);
- . the bit allocation for each sub-band within a frame;
- . the scale factors for each sub-band within each sub-block within a frame.

When editing decoded audio, improved quality can be obtained if additional data are sent to describe any timing offset that could have been introduced by editing only at points corresponding

to block boundaries (for example, at 48 kHz, 24-ms boundaries for MPEG-1 Layer II or MPEG-2 Layer II). This is particularly relevant for editing sound that has a close temporal relationship to accompanying pictures when loss of lip-movement synchronization is a problem.

This data signal can be used by a downstream encoder to guide its coding decisions. If the audio signal is changed in any way (for example, by mixing with another signal), then, in general, it would not be appropriate for the downstream encoder to re-use any previous coding decisions. However, in this case, if a suitable transport mechanism is used, the coding decision data become corrupted and the coder is able to detect that the data are no longer valid.

In this standard, it is assumed that audio coder control data are added to the least significant bit of the PCM audio signal in systems that handle the PCM audio with one of three different widths of PCM sample, namely, 16-bit, 20-bit, and 24-bit samples. In order to reduce the possibility of audibility of the coder-control data, the data are signaled by adjusting the parity of each 16-bit (20- or 24-bit) word appropriately. A CRC word is appended to frames of data to provide a means to detect errors in the data.

If the audio signal is changed in any way (for example, by mixing with another signal), then it is not appropriate that the downstream encoder re-use any previous coding decisions. However, in this case, the coder control data become corrupted and the coder is able to detect that the data are no longer valid.

#### 4 Recoding\_data for MPEG-1, or MPEG-2 low sampling frequency

The data are for MPEG-1 Layer II or for MPEG-2 LSF Layer II, where:

**ID** is the algorithm ID, according to ISO/IEC 11172-3, that is set to logic 1 to indicate ISO/IEC 11172-3, and set to logic 0 to indicate extension to lower sampling frequencies specified in ISO/IEC 13818-3;

**bitrate\_index** indicates the bit rate, according to ISO/IEC 11172-3;

**sampling\_frequency** indicates the sampling frequency, according to ISO/IEC 11172-3;

**mode** indicates the mode (monophonic, stereo, joint stereo, and so on), according to ISO/IEC 11172-3;

**mode\_extension** is used in joint stereo mode, according to ISO/IEC 11172-3;

**copyright** indicates copyright protection according to ISO/IEC 11172-3;

**original/copy** indicates whether copy is original or not, according to ISO/IEC 11172-3;

**emphasis** indicates the type of de-emphasis to use, according to ISO/IEC 11172-3;

**allocation** is the bit allocation index into the quantization table which shall be sent as a 4-bit integer (not a 2-, 3-, or 4-bit integer as would be used in the MPEG bit stream), with unused bits set to 0 and bit allocations for sub-bands that were not transmitted in the original coded bit stream set to 0; **[note “.” has been changed to “;”]**

**scalefactors** are factors represented by a 6-bit integer scale-factor index which should be transmitted in the audio coder control data, even if the original bit stream

used scale-factor select information to reduce the number sent, while scale factors for sub-bands that were not transmitted in the original coded bit stream should be set to 6210;

**padding\_bit** indicates bits that are added to bring the total number of bits in the recoding\_data to a multiple of 8 to match the length field.

## 5 Sample alignment of audio coder control data with respect to decoded-audio signal

Frame boundary positions in the decoded PCM audio can be determined from the position of the first bit of the audio coder control data. The relationship between the boundary and the position of this first bit depends on the coding scheme that was last used.

### 5.1 MPEG-1 Layer II or MPEG-2 Layer II

Because of the length of the response of the sub-band synthesis filters, when positioning the first coder control bit for MPEG-1 Layer II or MPEG-2 Layer II, a given frame of compressed data will give non-zero signal contributions extending beyond 24 ms (at a sampling frequency of 48 kHz) when decoded. However, the audio coder control data relating to a particular compressed audio frame may be carried on the PCM samples that are predominantly reconstructed from that particular frame (for example, for editing).

Therefore, the coder control data for a particular audio frame should be added to the decoded signal at the output of the synthesis filter bank after a delay corresponding to the delay through the filter bank. This delay has been measured to be 240 sample periods.

An example showing the relationship between coded data and decoded data for typical MPEG-1 Layer II or MPEG-2 Layer II codecs is included in annex B.

A method of transporting LSF (MPEG-2) signals on an AES3 interface, together with the audio coder control data signal, is described in **annex C**.

The LSF samples produced by the decoder shall be present, unmodified, in every second AES3 frame, when transported on an AES3 interface. In this case, the position of the first coder control bit shall indicate an unmodified LSF sample.

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## **Annex A (informative) to PART 2**

### **Use of the audio coder control data in a MPEG-1 Layer II or a MPEG-2 LSF Layer II environment**

## **Annex B (informative) to PART 2**

### **Alignment of MPEG-1 Layer II or of MPEG-2 LSF Layer II audio coder control data to decoded-audio samples**

## **Annex C (informative) to PART 2**

### **Transmission of audio coder control data signal associated with MPEG-2 LSF signals**

# AES41

## PART 3 MPEG-4 AAC and HE-AAC minimal metadata set

### 0 Introduction

*AES41 provides for the carriage of audio meta-data by embedding it in the audio samples themselves. This tightly associates the meta-data with the audio, yet makes it fragile so that changes to the audio will invalidate the meta-data. Several meta-data sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.*

This part describes the format for the data to be transmitted with audio to signal down-mix coefficients, loudness and dynamic range control meta-data as used in ISO/IEC 14496-3:2009 Subpart 4 (AAC and HE-AAC). The presence of this data in the stream is indicated by **type**, as defined in Part 1 of this standard, equal to 4.

#### 0.1 Rationale for part 3 of this standard.

Unwanted loudness variations in broadcast audio have historically been the source of many audience complaints. Broadcasters have adopted numerous techniques to address the problem – one of which involves indicating the long-term average loudness of the audio.

MPEG-4 AAC and HE-AAC are increasingly being used for broadcast sound services associated with standard- and high-definition television services, in stereo and surround, and the MPEG-4 standard makes provision for meta-data to control loudness and down-mix. This part of the AES41 standard enable dynamic meta-data to be carried to an MPEG-4 audio encoder as part of the PCM audio signal.

### 3.1 ISO/IEC 14496-3:2009 minimal metadata set

The data are a minimal subset of the ISO/IEC 14496-3 metadata, where

**matrix\_mixdown\_idx\_present** indicates whether the following **matrix\_mixdown\_idx** is valid

|   |                                    |
|---|------------------------------------|
| 0 | not valid, value should be ignored |
| 1 | valid                              |

**matrix\_mixdown\_idx** indicates the coefficient “A” as used in the ISO/IEC 14496-3:2009

matrix-mixdown process:

|   |         |          |
|---|---------|----------|
| 0 | 1/ 2    | (-3 dB)  |
| 1 | 1/2     | (-6 dB)  |
| 2 | 1/(2 2) | (-9 dB)  |
| 3 | 0       | (- ∞ dB) |

**prog\_ref\_level\_present** indicates whether the following **prog\_ref\_level** is valid

|   |                                    |
|---|------------------------------------|
| 0 | not valid, value should be ignored |
|---|------------------------------------|

1        valid

**prog\_ref\_level**        indicates program headroom relative to full scale as in

ISO/IEC 14496-3:2009, where 0 means the program is full scale and each step is 0,25 dB.

**dyn\_rng\_sgn** sign bit of the dyn\_rng\_ctl dynamic range control word

**dyn\_rng\_ctl** dynamic range control word, quantised in 0,25-dB steps, as in ISO/IEC 14496-3:2009

### **3.2 Sample alignment of data with respect to audio signal**

Frame boundary positions in the decoded PCM audio can be determined from the position of the first bit of the audio coder control data. The relationship between the boundary and the position of this first bit depends on the coding scheme that was last used. Because the anticipated use of this meta-data is *before* MPEG-4 audio coding rather than after, a frame boundary position has not yet been defined. Should it be deemed necessary, later versions of this standard may do so.

#### **3.2.1 ISO/IEC 14496-3:2009**

When the audio coder control data is the ISO/IEC 14496-3:2009 minimal data set, and the audio has not been previously encoded, then the first bit of the audio coder control data shall be embedded in the first audio sample to which it applies. The repetition rate of unchanging audio coder control data may be chosen to suit the requirements for “break-in” at arbitrary time to the audio signal of the application. At one extreme, this would be signaling only once, at the beginning of the stream. More generally, in anticipation of coding, a repetition every 1024 or 960 samples, corresponding to frame lengths in ISO/IEC 14496-3:2009 is recommended.

# AES 41

## PART 4 Dolby E minimal meta-data set

### 0 Introduction

*AES41 provides for the carriage of audio meta-data by embedding it in the audio samples themselves. This tightly associates the meta-data with the audio, yet makes it fragile so that changes to the audio will invalidate the meta-data. Several meta-data sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.*

This part describes the format for the data to be transmitted with audio to signal down-mix coefficients, loudness, and channel configuration meta-data as used Dolby E. The presence of this data in the stream is indicated by **type**, as defined in Part 1 of this standard, equal to 5.

### 0.1 Rationale for part 4 of this standard.

Unwanted loudness variations in broadcast audio have historically been the source of many audience complaints. Broadcasters have adopted numerous techniques to address the problem – one of which involves indicating the long-term average loudness of the audio. Dolby E is a technology developed to convey multi-channel audio using the AES3 digital audio interface. Multi-channel infra-structure is now much more commonly available, but while the requirement for Dolby E is declining, the need for the meta-data persists. Conveying the meta-data as part of the audio that it describes offers significant advantages over mechanisms such as embedding in vertical ancillary data in a video stream.

## 1 Dolby E audio minimal metadata

The data are a minimal subset of the Dolby E audio metadata, where:

**description\_text** multi-character text description of the associated programme. The value 0216 (STX) indicates that the next character is the start of the text description, while the value 0316 (ETX) indicates that the previous character was the end of the text description. The value 0016 (NUL) indicates that no text description information is present. Values between 2016 and 7E16 convey the corresponding ASCII character. All other values are reserved.

**acmod** indicates which channels are in use:

000 reserved

001 center

010 left, right

011 left, center, right

100 left, right, surround

101 left, center, right, surround

110 left, right, left surround, right surround

111 left, center, right, left surround, right surround

**cmixlev** indicates the down-mix level of the center channel with respect to the left and right

channels:

00 -3 dB

01 -4,5 dB

10 -6 dB

11 reserved

surmixlev indicates the down-mix level of the surround channels with respect to the left and right channels:

00 -3 dB

01 -4,5 dB

10 -6 dB

11 reserved

lfeon indicates that the low-frequency effects channel is on when '1', that it is off when '0'

dialnorm sign-inverted loudness value.

EXAMPLE: if loudness is -23 LKFS, the dialnorm value is 23.

# AES41

## PART 5 EBU loudness meta-data

### 0 Introduction

*AES41 provides for the carriage of audio meta-data by embedding it in the audio samples themselves. This tightly associates the meta-data with the audio, yet makes it fragile so that changes to the audio will invalidate the meta-data. Several meta-data sets have been defined, covering applications such as cascaded compression (bit rate reduction), and loudness control.*

This part describes the format for the data to be transmitted with audio to signal loudness and true peak meta-data as used in EBU Technical Recommendation R128. The presence of this data in the stream is indicated by **type**, as defined in Part 1 of this standard, equal to 6.

#### 0.1 Rationale for part 5 of this standard.

Unwanted loudness variations in broadcast audio have historically been the source of many audience complaints. Broadcasters have adopted numerous techniques to address the problem – one of which involves indicating the long-term average loudness of the audio.

The EBU studied the needs of audio signal levels in production, distribution and transmission of broadcast programmes. It formed the opinion that an audio-levelling paradigm was needed based on loudness measurement according to ITU-R BS.1770. This is described in EBU Technical Recommendation R 128, and supporting Technical Documents.

When the PCM signals form part of a 5-channel surround sound signal it may be necessary to include down-mix coefficients to allow later mixing of the center and surround channels into the front left and front right to form a two-channel stereo signal.

### 1 EBU Loudness meta-data

| Syntax                                                                                                                                                                                                                                                                                                 | No. of bits                                               | Mnemonic                                                                                  |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-------------------------------------------------------------------------------------------|
| recoding_data()<br>{<br>if(type==EBU loudness metadata){<br><b>LoudnessValue</b><br><b>LoudnessRange</b><br><b>MaxTruePeakLevel</b><br><b>MaxMomentaryLoudness</b><br><b>MaxShortTermLoudness</b><br><br><b>MPEG2_mix_level_on</b><br><b>Center_mix_level_value</b><br><b>Surround_mix_level_value</b> | <br><br><br>16<br>16<br>16<br>16<br>16<br><br>1<br>3<br>3 | <br><br><br>simsbf<br>simsbf<br>simsbf<br>simsbf<br>simsbf<br><br>bslbf<br>bslbf<br>bslbf |

|   |  |  |
|---|--|--|
| } |  |  |
|---|--|--|

The loudness parameters are represented by integer values, but preserve a precision of 2 decimal places by being multiplied by 100 before being rounded. The rounding function that shall be used is defined as follows:

integer representation,  $round(x) = \text{integer part of } (x + \text{sgn}(x) \cdot 0.5)$

where x is the loudness parameter to be represented, multiplied by 100, and  $\text{sgn}(x)$  is the signum function (-1 for  $x < 0$ ; 0 for  $x = 0$ ; +1 for  $x > 0$ ). This rounding method is commonly referred to as, "round to nearest, ties away from zero". Examples are shown in table XXX1

Table XXX1 examples of rounding and scaling method

| Parameter value | Integer representation | Parameter value | Integer representation |
|-----------------|------------------------|-----------------|------------------------|
| 12.766          | 1277                   | -22.644         | -2264                  |
| 12.765          | 1277                   | -22.645         | -2265                  |
| 12.764          | 1276                   | -22.646         | -2265                  |

|                                 |                                                                                                                                                                                                                   |
|---------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>LoudnessValue</b>            | A 16-bit signed integer, equal to $round(100 \times \text{the Integrated Loudness Value of the signal in LUFS})$                                                                                                  |
| <b>LoudnessRange</b>            | 16-bit signed integer, equal to $round(100 \times \text{the Loudness Range of the signal in LU})$                                                                                                                 |
| <b>MaxTruePeakLevel</b>         | 16-bit signed integer, equal to $round(100 \times \text{the Maximum True Peak Value of the signal in dBFS})$                                                                                                      |
| <b>MaxMomentaryLoudness</b>     | 16-bit signed integer, equal to $round(100 \times \text{the highest value of the Momentary Loudness Level of the signal in LUFS})$                                                                                |
| <b>MaxShortTermLoudness</b>     | 16-bit signed integer, equal to $round(100 \times \text{the highest value of the Short-term Loudness Level of the signal in LUFS})$                                                                               |
| <b>MPEG2_mix_level_on</b>       | Indicates whether the following <b>mix_level_value</b> parameters are valid<br>0 not valid, value should be ignored<br>1 valid                                                                                    |
| <b>Center_mix_level_value</b>   | Indicates the level at which the center channel shall be mixed into front left and front right. Corresponding multiplication factors are shown in table XXX                                                       |
| <b>Surround_mix_level_value</b> | Indicates the level at which the left surround and right surround channels shall be mixed into the front left and front right channels, respectively. Corresponding multiplication factors are shown in table XXX |

Note 1: The loudness terms are explained more fully in EBU Recommendation R 128 [?], EBU Tech Doc 3341 [?] and EBU Tech Doc 3342 [?], supported by EBU Tech Doc 3343 and EBU Tech Doc 3344.

Note 2: Not all loudness metadata fields will be measured or required in all circumstances. The value -1 ('0111 1111 1111 1111' as a bit string) shall be used to signal “field not indicated”. Parameters signalled in this way shall be ignored.

Values of loudness level or true peak that are signalled outside the range -99.9 to +99.9 are invalid and shall be ignored. Values of loudness range that are signalled outside the range 0.00 to +99.9 are invalid and shall be ignored.

Table XXX mix\_level\_value multiplication factors

| Bit string | Multiplication factor | Equivalent gain |
|------------|-----------------------|-----------------|
| "000"      | 1.000                 | 0.0 dB          |
| "001"      | 0.841                 | -1.5 dB         |
| "010"      | 0.707                 | -3.0 dB         |
| "011"      | 0.596                 | -4.5 dB         |
| "100"      | 0.500                 | -6.0 dB         |
| "101"      | 0.422                 | -7.5 dB         |
| "110"      | 0.355                 | -9.0 dB         |
| "111"      | 0.000                 | $-\infty$ dB    |